

DEVOPS SOP – Fusionpact Gauntlet Challenge

 **Name: Aditya Rauniyar**

Project Title:

Dockerized Full-Stack Application with CI/CD & Monitoring (Prometheus + Grafana)

GitHub Repository:

<https://github.com/adityarauniyar-ux/fusionpact-devops-challenge>

1. Project Overview

This project is a complete **DevOps implementation** of a full-stack application consisting of a frontend service, backend API, and a monitoring stack.

The goal was to containerize services, automate deployment, and implement observability using industry-standard tools.

GitHub Repository:

<https://github.com/adityarauniyar-ux/fusionpact-devops-challenge>

2. Architecture Design

Frontend (Nginx)

↓

Backend (FastAPI - Python)

↓

/metrics endpoint (Prometheus client)

↓

Prometheus (Data scraping)

↓

Grafana (Dashboard visualization)

3. Docker Setup

All services are containerized using Docker to ensure consistency across environments.

Backend Dockerfile

FROM python:3.10

WORKDIR /app

COPY requirements.txt . RUN pip

install -r requirements.txt COPY . .

EXPOSE 8000

CMD ["uvicorn", "app.main:app", "--host", "0.0.0.0", "--port", "8000"]

Frontend (Nginx container)

- Static frontend served using Nginx
 - Port mapping: 3000 → 80
-

4. Docker Compose Orchestration

version: "3.8"

backend: build: ./backend

container_name:

backend_app

ports:

- "8000:8000"

restart: always

frontend:

build: ./frontend

container_name: frontend_app

ports: -

"3000:80"

depends_on: -

backend

restart:

always

prometheus:

image: prom/prometheus

container_name: prometheus volumes:

- ./prometheus.yml:/etc/prometheus/prometheus.yml **ports:**

- "9090:9090" restart: always

grafana:

image: grafana/grafana

container_name: grafana

ports:

- "3001:3000" restart: always

5. Monitoring Setup (Prometheus + Grafana)

1.Backend Metrics Implementation

A custom Prometheus metric was added to track request counts.

from fastapi import FastAPI, Response from

prometheus_client import Counter, generate_latest

app = FastAPI()

REQUEST_COUNT = Counter("app_requests_total", "Total API Requests")

@app.get("/")

def home():

REQUEST_COUNT.inc()

return {"message": "API is running successfully"}

@app.get("/metrics")

def metrics():

return Response(generate_latest(), media_type="text/plain")

2.Prometheus Configuration

global:

scrape_interval: 5s

scrape_configs:

- job_name: "backend_service" static_configs:

- targets: ["backend:8000"]

6. CI/CD Pipeline (GitHub Actions)

CI

Pipeline

on: push:

branches: [main

] jobs: build:

runs-on: ubuntu-latest

steps:

- name: Checkout

Repository uses:

actions/checkout@v3

- name: Build Docker Images run: docker-compose

build

- name: Start

Containers run: docker-compose up

-d

A simple CI pipeline was configured for automated build testing.

7. Data Persistence Strategy

Although the application is containerized and stateless, persistence is ensured through:

- Docker volume isolation for service independence
 - Prometheus local storage for metrics retention
 - Grafana dashboard configuration stored in container volumes
 - Backend designed to be stateless for scalability
-

8. Execution Commands

docker-compose up --build

docker ps docker logs

backend_app

git add .

git commit -m "Added full DevOps stack with monitoring and CI/CD"

git push origin main

9. Verification Steps

Backend API - <http://localhost:8000>

Metrics Endpoint - <http://localhost:8000/metrics>

Prometheus Dashboard - <http://localhost:9090>

Login: admin / admin

10. Key Learnings

- Containerization using Docker
 - Multi-service orchestration using Docker Compose
 - Observability using Prometheus + Grafana
 - CI/CD automation using GitHub Actions
 - Real-world DevOps workflow understanding
-

11. Live Links

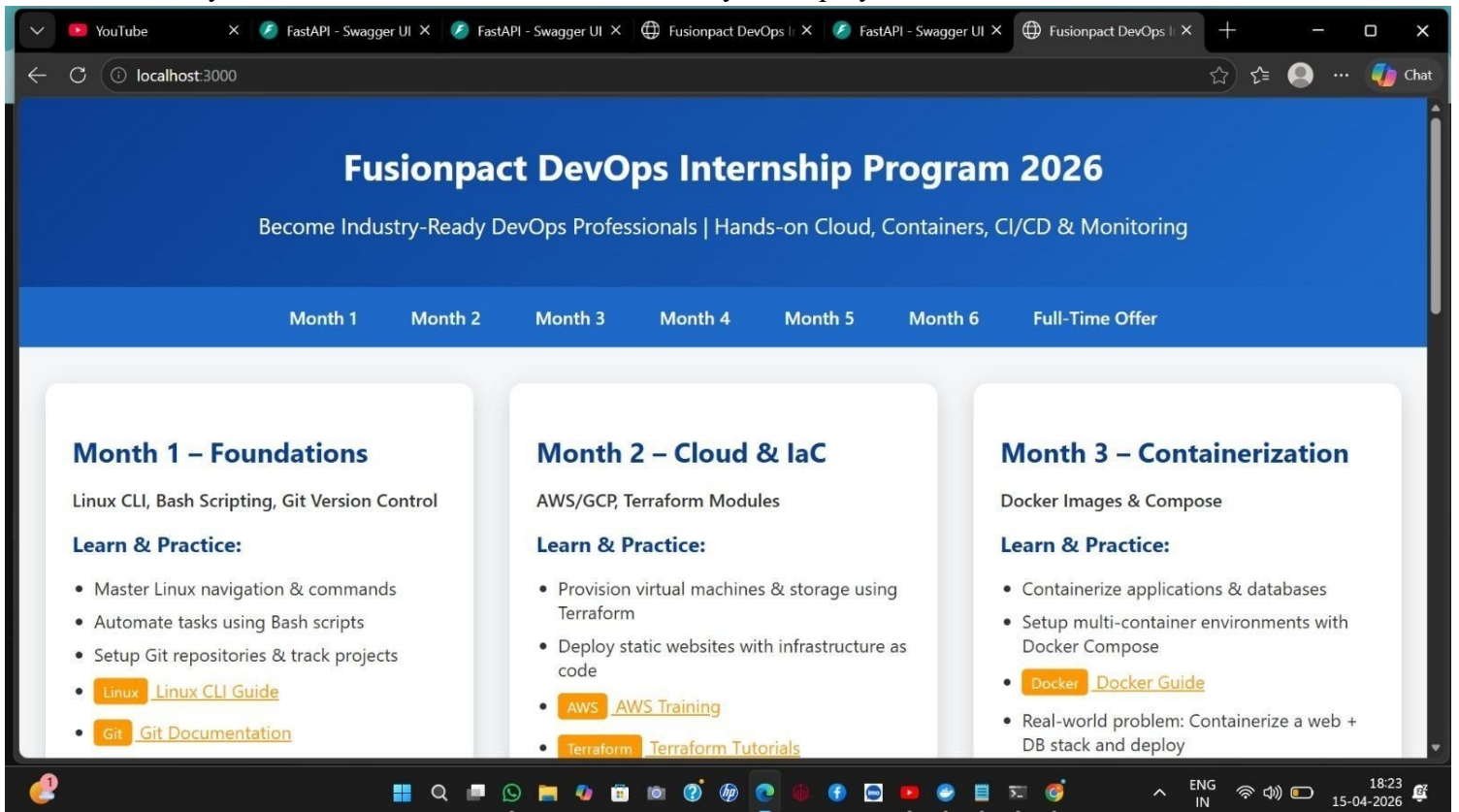
Frontend: <https://frontend-app-0qf9.onrender.com>

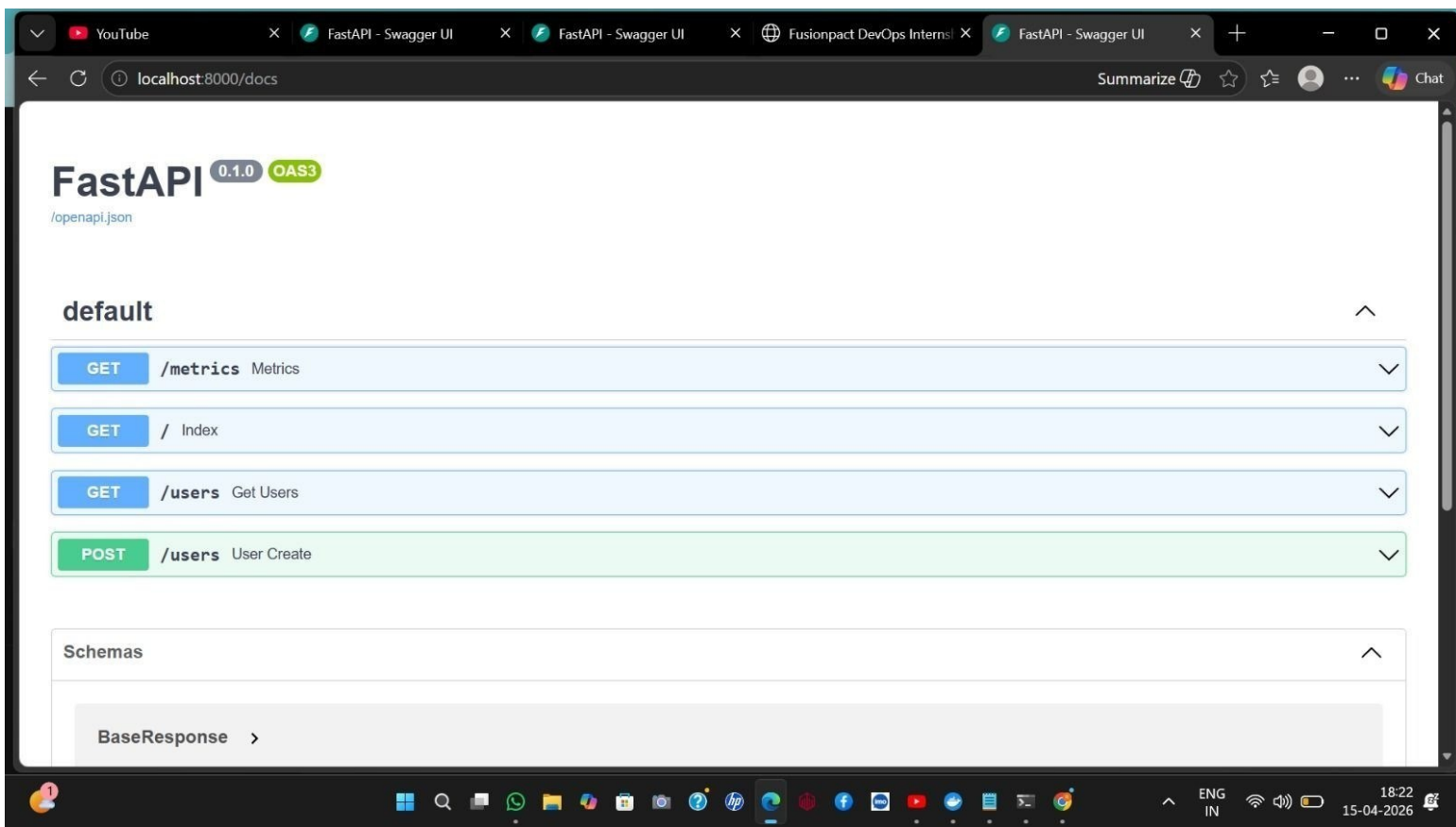
Backend: <https://fusionpack-devops-challenge-9mdt.onrender.com>

12. Final Summary

This project demonstrates a production-style DevOps pipeline with full-stack integration, monitoring, and automation.

The system is scalable, containerized, and ready for deployment in a real-world environment.





Docker Running Containers

```
Command Prompt
C:\Users\HP\fusionpact-devops-challenge\backend>docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
8fe074046959   fusionpact-devops-challenge-frontend  "/docker-entrypoint.    20 minutes ago Up About a minute 0.0.0.0:3000->80/
tcp, [::]:3000->80/tcp   frontend_app
7095caba2f0c   prom/prometheus                       "/bin/prometheus --c    20 minutes ago Up About a minute 0.0.0.0:9090->909
0/tcp, [::]:9090->9090/tcp prometheus
46847eed044c   grafana/grafana                       "/run.sh"               20 minutes ago Up About a minute 0.0.0.0:3001->300
0/tcp, [::]:3001->3000/tcp grafana
376c11064585   fusionpact-devops-challenge-backend  "uvicorn app.main:ap    20 minutes ago Up About a minute 0.0.0.0:8000->800
0/tcp, [::]:8000->8000/tcp backend_app
C:\Users\HP\fusionpact-devops-challenge\backend>
```

```
Command Prompt

C:\Users\HP\fusionpact-devops-challenge\frontend>docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
8fe074046959   fusionpact-devops-challenge-frontend  "/docker-entrypoint..." 22 minutes ago Up 3 minutes  0.0.0.0:3000->80/tcp,
[::]:3000->80/tcp   frontend_app
7095caba2f0c    prom/prometheus                      "/bin/prometheus --c..." 22 minutes ago Up 3 minutes  0.0.0.0:9090->9090/tcp
, [::]:9090->9090/tcp prometheus
46847eed044c    grafana/grafana                      "/run.sh"                22 minutes ago Up 3 minutes  0.0.0.0:3001->3000/tcp
, [::]:3001->3000/tcp grafana
376c11064585    fusionpact-devops-challenge-backend  "uvicorn app.main:ap..." 22 minutes ago Up 3 minutes  0.0.0.0:8000->8000/tcp
, [::]:8000->8000/tcp backend_app

C:\Users\HP\fusionpact-devops-challenge\frontend>
```

Command Execution

```
Command Prompt

C:\Users\HP\fusionpact-devops-challenge>dir
Volume in drive C is Windows
Volume Serial Number is 32A5-368C

Directory of C:\Users\HP\fusionpact-devops-challenge

15-04-2026  22:30    <DIR>          .
15-04-2026  17:49    <DIR>          ..
15-04-2026  21:25    <DIR>          .github
15-04-2026  17:49            45 .gitignore
15-04-2026  18:06    <DIR>          backend
15-04-2026  22:12           551 docker-compose.yml
15-04-2026  18:14    <DIR>          frontend
15-04-2026  22:30           130 prometheus.yml
15-04-2026  17:49        4,924 README.md
15-04-2026  17:49       754,629 SOP CREATE HOME WEBPAGE USING NGINX SERVER.pdf
               5 File(s)      760,279 bytes
               5 Dir(s)   3,732,557,824 bytes free

C:\Users\HP\fusionpact-devops-challenge>
```

```
Command Prompt
15-04-2026 17:49 <DIR>      ..
15-04-2026 21:25 <DIR>      .github
15-04-2026 17:49      45 .gitignore
15-04-2026 18:06 <DIR>      backend
15-04-2026 21:53      0 docker-compose.yml
15-04-2026 18:14 <DIR>      frontend
15-04-2026 17:49      4,924 README.md
15-04-2026 17:49      754,629 SOP CREATE HOME WEBPAGE USING NGINX SERVER.pdf
      4 File(s)          759,598 bytes
      5 Dir(s)      4,914,634,752 bytes free

C:\Users\HP\fusionpack-devops-challenge>notepad docker-compose.yml

C:\Users\HP\fusionpack-devops-challenge>dir
Volume in drive C is Windows
Volume Serial Number is 32A5-368C

Directory of C:\Users\HP\fusionpack-devops-challenge

15-04-2026 21:53 <DIR>      .
15-04-2026 17:49 <DIR>      ..
15-04-2026 21:25 <DIR>      .github
15-04-2026 17:49      45 .gitignore
15-04-2026 18:06 <DIR>      backend
15-04-2026 21:56      268 docker-compose.yml
15-04-2026 18:14 <DIR>      frontend
15-04-2026 17:49      4,924 README.md
15-04-2026 17:49      754,629 SOP CREATE HOME WEBPAGE USING NGINX SERVER.pdf
      4 File(s)          759,866 bytes
      5 Dir(s)      4,913,614,848 bytes free

C:\Users\HP\fusionpack-devops-challenge>
```

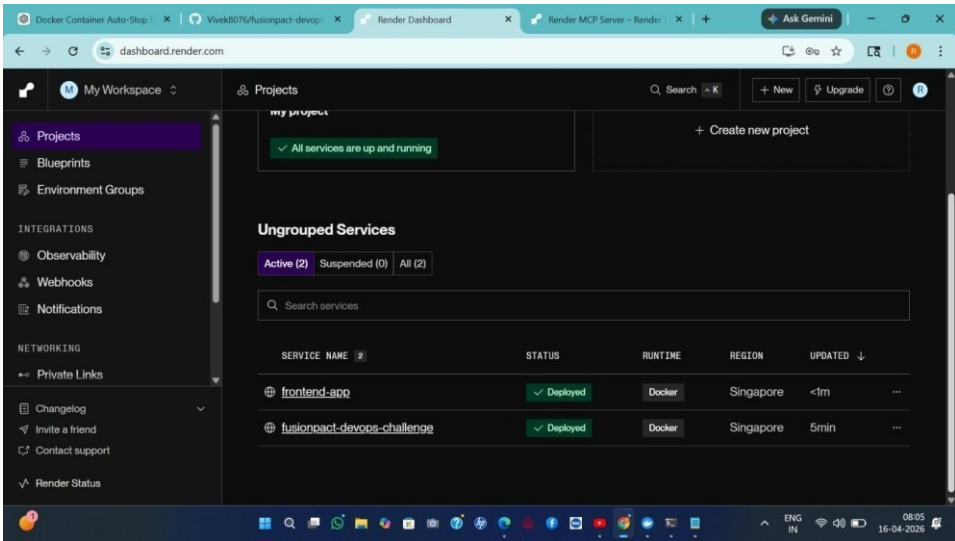
Command Execution

```
Command Prompt - docker-c
frontend_app | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
frontend_app | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
frontend_app | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
frontend_app | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
frontend_app | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
frontend_app | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
frontend_app | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
frontend_app | /docker-entrypoint.sh: Configuration complete; ready for start up
frontend_app | 2026/04/15 16:27:55 [notice] 1#1: using the "epoll" event method
frontend_app | 2026/04/15 16:27:55 [notice] 1#1: nginx/1.29.8
frontend_app | 2026/04/15 16:27:55 [notice] 1#1: built by gcc 15.2.0 (Alpine 15.2.0)
frontend_app | 2026/04/15 16:27:55 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
frontend_app | 2026/04/15 16:27:55 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
frontend_app | 2026/04/15 16:27:55 [notice] 1#1: start worker processes
frontend_app | 2026/04/15 16:27:55 [notice] 1#1: start worker process 30
frontend_app | 2026/04/15 16:27:55 [notice] 1#1: start worker process 31
frontend_app | 2026/04/15 16:27:55 [notice] 1#1: start worker process 32
frontend_app | 2026/04/15 16:27:55 [notice] 1#1: start worker process 33
frontend_app | 2026/04/15 16:27:55 [notice] 1#1: start worker process 34
frontend_app | 2026/04/15 16:27:55 [notice] 1#1: start worker process 35
frontend_app | 2026/04/15 16:27:55 [notice] 1#1: start worker process 36
frontend_app | 2026/04/15 16:27:55 [notice] 1#1: start worker process 37
frontend_app | 2026/04/15 16:27:55 [notice] 1#1: start worker process 38
frontend_app | 2026/04/15 16:27:55 [notice] 1#1: start worker process 39
frontend_app | 2026/04/15 16:27:55 [notice] 1#1: start worker process 40
frontend_app | 2026/04/15 16:27:55 [notice] 1#1: start worker process 41
backend_app  | INFO:      Started server process [1]
backend_app  | INFO:      Waiting for application startup.
backend_app  | INFO:      Application startup complete.
backend_app  | INFO:      Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
```

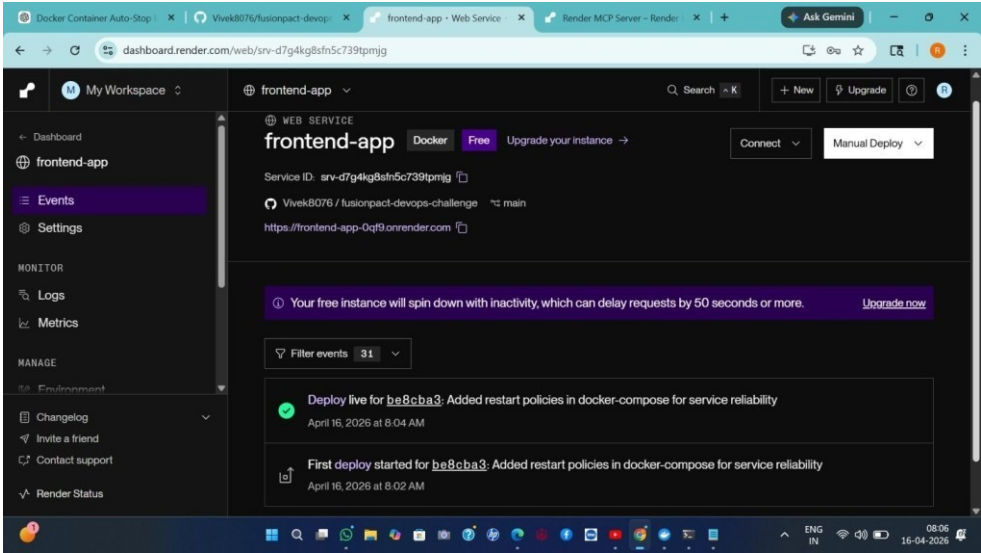

Screenshots

The following screenshots demonstrate the successful local execution and cloud deployment of the application

Render Dashboard



Frontend (Live Application)



Backend (API Service)

